

Dynamics and Control (B) Report (Part of Control)

BY

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(1) Consider a DC motor system which input is a voltage $u(t)$ and output is rotational angular velocity $y(t)$ of DC motor. Let $U(s)$ and $Y(s)$ are Laplace transform of input and output, respectively. Explain that transfer function $G(s)$ can be explained as following equation.

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K}{1 + sT}$$

Answer:

$$y(t) = \frac{d\theta(t)}{dt}$$

From Kirchhoff's voltage law,

$$u(t) = Ri(t) + K_E y(t) \quad (1)$$

From Rotational motion equation:

$$J \frac{dy(t)}{dt} + By(t) = K_T i(t) \quad (2)$$

Where:

- **R**: Armature Resistance
- **J**: Moment of inertia
- **B**: Viscous friction coefficient
- **K_E** : Back EMF constant
- **K_T** : Torque constant
- **θ** : Motor rotational angle

By taking Laplace transform (zero initial condition):

From equation (1):

$$U(s) = RI(s) + K_E Y(s) \quad (3)$$

From Equation (2):

$$(Js + B)Y(s) = K_T I(s)$$

$$I(s) = \frac{(Js + B)}{K_T} Y(s)$$

By substituting $I(s)$, equation (3) becomes:

$$U(s) = R \left[\frac{(Js + B)}{K_T} Y(s) \right] + K_E Y(s)$$

$$U(s) = \left(\frac{R(Js + B)}{K_T} + K_E \right) Y(s)$$

Thus, transfer function from input voltage to angular velocity is:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{Y(s)}{\left(\frac{R(Js + B)}{K_T} + K_E\right)Y(s)}$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{\left(\frac{R(Js + B)}{K_T} + K_E\right)}$$

$$G(s) = \frac{K_T}{(RJs + (RB + K_T K_E))}$$

$$G(s) = \frac{\frac{K_T}{RB + K_T K_E}}{\left(\frac{RJs}{RB + K_T K_E} + \frac{RB + K_T K_E}{RB + K_T K_E}\right)}$$

With $K(\text{gain}) = \frac{K_T}{BR + K_E K_T}$ and $T(\text{Time constant}) = \frac{JR}{BR + K_E K_T}$:

The equation for $G(s)$ becomes:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K}{1 + Ts}$$

(2) Find the unit step response of $y(t)$ of the DC motor shown in 1. Also, change the parameters K and T of the DC motor system into various values, draw a graph of the response behavior, and consider the meaning of the parameters K and T . (Try numerical calculation using Excel, C language, Python, etc.)

Answer:

For a unit step input:

$$u(t) = 1$$

By taking Laplace Transform:

$$U(s) = \frac{1}{s}$$

The output becomes:

$$Y(s) = G(s)U(s) = \frac{K}{s(1 + Ts)}$$

By simplifying, $Y(s)$ becomes:

$$Y(s) = \frac{K}{s} + \frac{-KT}{(1 + Ts)}$$

By taking inverse Laplace transform:

$$y(t) = K(1 - e^{-t/T}), t \geq 0$$

This is the unit-step response of the DC motor angular velocity.

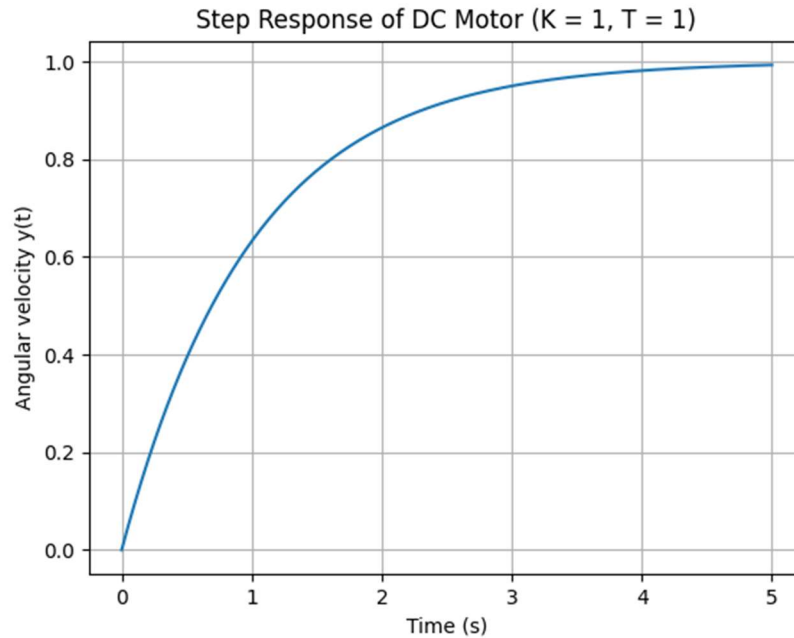


Figure 1. DC motor output response for $K = 1$ and $T = 1$

Figure 1 is the DC motor output response with the $K = 1$ and $T = 1$. The curve shows a standard first-order response where the angular velocity increases smoothly from zero and approaches the steady-state condition. The response has no oscillation, and the angular velocity reaches approximately 65% of its final value in 1 s.

Figure 2 illustrates the step responses when the gain K is fixed and the time constant T is varied. In this case, all response curves converge to the same steady-state value because the gain K remains constant. However, the speed of the response changes significantly with different values of T . Smaller values of T result in faster rise times and quicker convergence to the steady-state value, while larger values of T produce slower responses. This demonstrates that the time constant T controls the dynamic speed of the system without affecting the final angular velocity.

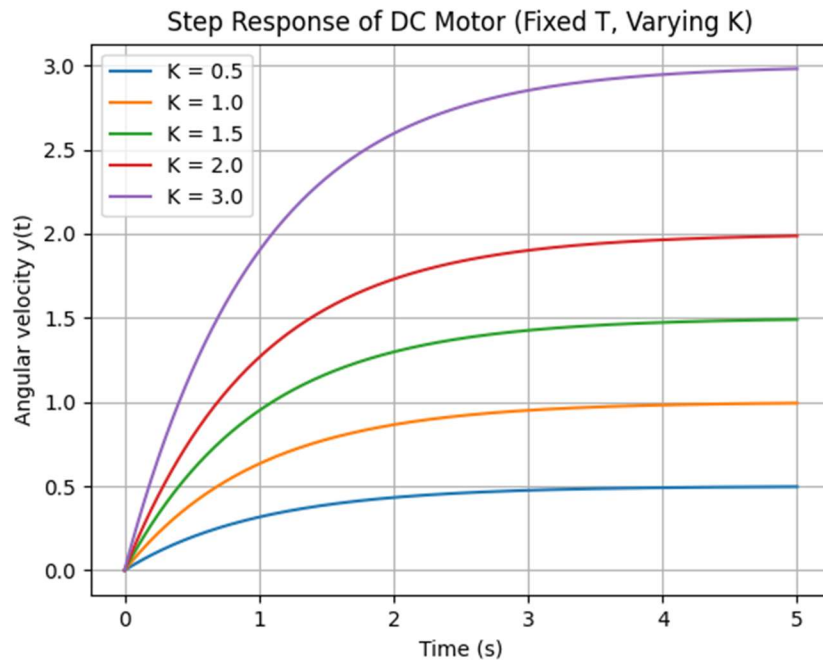


Figure 2. DC motor output response for Fixed T and Varying K

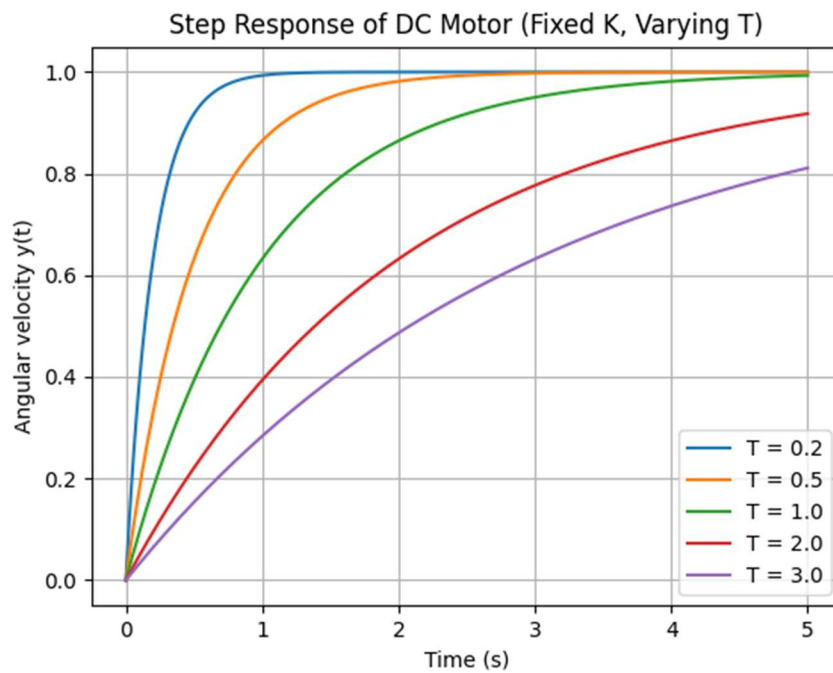


Figure 3. DC motor output response for Fixed K and Varying T

Figure 3 presents the step responses when the time constant T is fixed and the gain K is varied. In this case, all responses exhibit the same rise speed because the time constant remains unchanged. However, the steady-state value of the angular velocity differs for each curve and

increases proportionally with K . This shows that the gain K determines the magnitude of the steady-state response, while it does not influence the response speed. From this graph, it can be concluded that the parameters K and T independently control the steady-state and transient characteristics of the DC motor system.

The parameter K in the transfer function $G(s) = \frac{K}{1+Ts}$ represents the steady-state gain of the DC motor system. It determines the final angular velocity reached by the motor when a constant voltage input is applied. For a unit step input, the steady-state value of the output is equal to K , meaning that a larger value of K results in a higher final rotational speed. Physically, K is influenced by the motor torque constant, the back electromotive force constant, the electrical resistance, and the mechanical friction of the motor. Therefore, K reflects the efficiency of the motor in converting electrical energy into mechanical motion and controls the magnitude of the steady-state response.

The parameter T is the time constant of the DC motor system and characterizes the speed of the transient response. It indicates how quickly the angular velocity increases from zero to its steady-state value after a voltage is applied. A smaller value of T corresponds to a faster response, while a larger value of T results in a slower rise in output. For a unit-step input, the angular velocity reaches approximately 63.2% of its final value at time $t = T$, which provides a clear physical interpretation of the time constant. The value of T depends mainly on the moment of inertia of the motor and the damping effects in the system, and it governs the dynamic behavior of the motor without affecting the steady-state value.

Appendix:

Python script for the evaluation of $y(t)$ with $K = 1$ and $T = 1$.

```
import numpy as np
import matplotlib.pyplot as plt

# Time axis
t = np.linspace(0, 5, 500)

# Parameters
K = 1.0
T = 1.0
# Unit step response
y = K * (1 - np.exp(-t / T))
# Plot
plt.figure()
plt.plot(t, y)
plt.xlabel("Time (s)")
plt.ylabel("Angular velocity y(t)")
plt.title("Step Response of DC Motor (K = 1, T = 1)")
plt.grid()
plt.show()
```

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